11) Prove or disprove that the product of two irrational numbers is irrational.

Soln: We have already proved that $\sqrt{2}$ is an irrational no.

$(\sqrt{2})^2 = \sqrt{2} \cdot \sqrt{2} = 2 = \frac{2}{1}$. By defn. 2, 2 is a rational no.

We have seen that the product of two irrational no.s is rational.

$\therefore$ We have disproved that the product of two irrational numbers is irrational by a counterexample. $\square$

12) Prove or disprove that the product of a nonzero rational number and an irrational number is irrational.

Soln: p: x is an nonzero rational number q: y is an irrational number

r: xy is irrational

We shall prove $(p \wedge q) \rightarrow r$ is True by contradiction

We shall prove $(p \wedge q) \wedge \neg r \rightarrow F$ is True

$\therefore$ We shall prove $\boxed{(p \wedge q) \wedge \neg r} \rightarrow F$ is True

Let, $(p \wedge q) \wedge \neg r$ be True. By defn. of AND, $(p \wedge q)$ is True and $\neg r$ is True.

$\because \neg r$ is True, $\therefore xy$ is not irrational, i.e. xy is rational (Defn. 2)

By defn of AND, p is True, q is True.

$\therefore x$ is an nonzero rational number. By defn 2, $\exists$ integers a, b where $b \neq 0$ s.t. $x = \frac{a}{b}$. $\because x \neq 0$, $\therefore a \neq 0$.

By defn 2, $\exists$ integers c, d where $d \neq 0$ s.t. $xy = \frac{c}{d}$

$y = \frac{xy}{x} = \frac{c/d}{a/b} = \frac{cb}{ad}$. Its easy to see that cb, ad are both integers.

$\because a \neq 0, d \neq 0$ $\therefore ad \neq 0$. By defn 2, y is a rational number. This is a contradiction

$\therefore \neg r$ is False. $\therefore$ By defn. of AND, $(p \wedge q) \wedge \neg r$ is ~~True~~ False. By defn. of conditional, $\boxed{(p \wedge q) \wedge \neg r} \rightarrow F$ is True. $\therefore (p \wedge q) \rightarrow r$ is True.